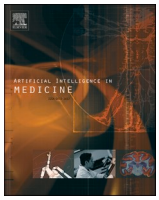
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Research paper



Deep learning algorithms for melanoma detection using dermoscopic images: A systematic review and meta-analysis☆

Zichen Ye [a], Daqian Zhang [a], Yuankai Zhao [a], Mingyang Chen [a], Huike Wang [a], Samuel Seery [b], Yimin Qu [a], Peng Xue [a,*], Yu Jiang [a,*]

[a] School of Population Medicine and Public Health, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China
[b] Population Health Sciences Institute, School of Pharmacy, Newcastle University, Newcastle NE1 7RU, United Kingdom of Great Britain and Northern Ireland

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ABSTRACT

Background: Melanoma is a serious risk to human health and early identification is vital for treatment success. Deep learning (DL) has the potential to detect cancer using imaging technologies and many studies provide evidence that DL algorithms can achieve high accuracy in melanoma diagnostics.
Objectives: To critically assess different DL performances in diagnosing melanoma using dermatoscopic images and discuss the relationship between dermatologists and DL.
Methods: Ovid-Medline, Embase, IEEE Xplore, and the Cochrane Library were systematically searched from inception until 7th December 2021. Studies that reported diagnostic DL model performances in detecting melanoma using dermatoscopic images were included if they had specific outcomes and histopathologic confirmation. Binary diagnostic accuracy data and contingency tables were extracted to analyze outcomes of interest, which included sensitivity (SEN), specificity (SPE), and area under the curve (AUC). Subgroup analyses were performed according to human-machine comparison and cooperation. The study was registered in PROSPERO, CRD42022367824.
Results: 2309 records were initially retrieved, of which 37 studies met our inclusion criteria, and 27 provided sufficient data for meta-analytical synthesis. The pooled SEN was 82 % (range 77–86), SPE was 87 % (range 84–90), with an AUC of 0.92 (range 0.89–0.94). Human-machine comparison had pooled AUCs of 0.87 (0.84–0.90) and 0.83 (0.79–0.86) for DL and dermatologists, respectively. Pooled AUCs were 0.90 (0.87–0.93), 0.80 (0.76–0.83), and 0.88 (0.85–0.91) for DL, and junior and senior dermatologists, respectively. Analyses of human-machine cooperation were 0.88 (0.85–0.91) for DL, 0.76 (0.72–0.79) for unassisted, and 0.87 (0.84–0.90) for DL-assisted dermatologists.
Conclusions: Evidence suggests that DL algorithms are as accurate as senior dermatologists in melanoma diagnostics. Therefore, DL could be used to support dermatologists in diagnostic decision-making. Although, further high-quality, large-scale multicenter studies are required to address the specific challenges associated with medical AI-based diagnostics.

1. Introduction

Skin cancers are malignant tumors that are highly prevalent in some parts of the world and can be life-threatening [1]. Among the types of skin cancer, melanoma has the highest rate of mortality and is responsible for 75 % of deaths related to skin tumors [2]. According to recent statistics in Cancer Today, there were approximately 331,722 new melanoma cases and around 58,667 deaths worldwide in 2022 [3]. As with all forms of cancer, early accurate diagnosis is key. Fortunately, a dermatoscope, used in skin cancer screening, is an efficient, noninvasive diagnostic tool, which has good auxiliary functions and can ensure early identification [4–7]. Histopathologic examination is the gold standard for melanoma detection [8]; however, there are issues related to operating skills and experience [9–11], and there is a global shortage of dermatologists [11,12]. These issues are compounded and further complicated by the fact that melanoma is often diagnosed late [13].

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* Corresponding authors.
E-mail addresses: xuepeng_pumc@foxmail.com (P. Xue), jiangyu@pumc.edu.cn (Y. Jiang).

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Deep learning (DL) is an artificial intelligence (AI) technique that discerns patterns and features through extensive data learning [14]. By using multi-layered neural networks, DL automatically extracts and assimilates complex data characteristics, enabling the identification and classification of patterns and variations [15,16]. In melanoma detection using dermoscopic image analysis, DL helps identify subtle image variations, improving both diagnostic accuracy and speed. Essentially, DL enables computers to solve complex problems by mimicking the data processing and analysis methods of the human brain. Therefore, DL algorithms provide a potential solution to address the challenges during dermatoscopic diagnosis. DL algorithms have performed well in several automatic detection but they must undergo validation before implementation [17,18]. There appears to be potential for DL algorithms in detecting melanoma, [19–24]. Indeed, many studies have reported that DL performs on par with, and sometimes even surpasses dermatologists. However, there is a need to synthesize evidence for policy development and to ensure the most effective changes are implemented correctly.

Several previous reviews have discussed barriers to the application of AI in dermoscopy [25], AI techniques for skin cancer detection and classification [26], and the diagnostic performance of convolutional neural networks (CNN) in melanoma [27], along with CNN's performance in classifying skin lesions [28]. However, these studies have mainly focused on qualitative aspects, and there is no comprehensive quantitative evaluation of AI diagnostic performance. In particular, a study that synthesizes the best available evidence in this field is absent. In fact, since Rajpara et al.'s (2009) meta-analysis which assessed the diagnostic accuracy of dermoscopy and digital dermoscopy/AI diagnosis of melanoma [29], few have systematically studied the performance of DL in melanoma diagnosis. Therefore, it is necessary to quantitatively synthesize DL performances in melanoma diagnosis with human-machine comparison. This quantitative approach could provide more reliable evidence for clinical practice, helping physicians and patients to better understand and accept the application of AI technologies.

2. Method

2.1. Protocol registration and study design

The study protocol was registered with the PROSPERO International Register of Systematic Reviews (reference CRD42022367824). This study is also reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines [30]. A completed PRISMA checklist has been provided as Supplementary file 1. No ethical approval or informed consent was required for this secondary data analysis.

2.2. Search strategy and eligibility criteria

Ovid-Medline, Embase, IEEE Xplore, and the Cochrane Library were systematically searched until 7th December 2021, with no language, regional, or publication restrictions. The full search strategy for each database was developed in collaboration with a group of experienced clinicians and medical researchers. Please refer to Supplementary file 2 for further details.

Articles screening was considered by two independent authors according to the titles and abstracts before full-texts were read. Discrepancies were resolved through discussion with a third collaborator. Studies that reported diagnostic performances of DL models for the detection of melanoma using dermatoscopic images were included.

Exclusion criteria included: (1) diagnosis of other non-melanoma skin tumors; (2) studies with no dermatoscopic images; (3) studies without histopathologic confirmation as the reference standard; and (4) non-DL research. Case reports, reviews, editorials, letters, commentaries, and conference abstracts/proceedings were also excluded.

2.3. Data extraction

Study characteristics and diagnostic performance data were independently extracted by two researchers using a predefined data extraction table. Again, differences were resolved by a third researcher. Binary diagnostic accuracy data were directly extracted into contingency tables, and categorical diagnostic accuracy data were converted into binary diagnostic accuracy data, namely melanoma and non-melanoma, including true-positives, false-positives, true-negatives, and false-negatives. These were then used to calculate pooled sensitivity (SEN), pooled specificity (SPE), and other metrics. If certain studies provided multiple contingency tables for the same or different DL algorithms, we assumed they were independent.

2.4. Quality assessment

The risk of bias and applicability of the included studies were quality assessed using customized assessment criteria based on the Quality Assessment tool of Diagnostic Accuracy Studies 2 (QUADAS-2) [31], QUADAS-Comparative (QUADAS-C) [32] and the quality assessment tool for artificial intelligence-centered diagnostic test accuracy studies (QUADAS-AI) [33,34] by two independent authors. Conflicts were discussed with a third author. Further details have been provided in Supplementary file 2.

2.5. Statistical analysis

Hierarchical summary receiver operating characteristic (SROC) curves were fitted to assess diagnostic DL algorithm performances. 95 % confidence intervals (CI) and predicted areas generated averaged SEN, SPE, and area under the curves (AUCs) were estimated in SROC figures. Meta-analysis was conducted to identify the highest diagnostic accuracy using multiple DL algorithms.

Heterogeneity was assessed using Spearman's correlation coefficient and I^2 statistics. Spearman's correlation coefficient test was used to assess the presence of a threshold effect. I^2 tests were used to consider non-threshold effects, and a P of > 0.05 or $I^2 > 50$ % provided evidence of substantial between-study heterogeneity. Subgroup and regression analyses were conducted to explore potential sources of heterogeneity. The random effects model was implemented because it was assumed that these studies were methodologically dissimilar. Publication bias was also assessed intuitively using a funnel plot.

Three subgroup analyses were conducted: (1) human-machine comparison (i.e. DL models versus dermatologists); (2) human-machine comparison with different seniority (i.e. junior and senior dermatologists); and (3) human-machine cooperation (i.e. DL vs unassisted dermatologists and DL-assisted dermatologists). In each subgroup, data were calculated using the DL algorithm with the highest diagnostic accuracy in each study.

Meta-analysis was only conducted when there were four or more original studies. Review Manager (version 5.4) and Stata (version 17.0) were used for quality evaluation plots and data analysis respectively. Data synthesis was performed within the bivariate mixed-effects binary regression modeling framework using the Midas package. The threshold for statistical significance was set to $P < 0.05$, and all tests were two-tailed.

3. Results

3.1. Study selection and characteristics of eligible studies

The initial search identified 2309 records, of which 235 duplicates were removed using referencing software. A further 1848 studies were removed according to titles and abstracts. 226 records on skin tumors were included, of which six could not be retrieved, and after reading 220 full-texts a further 183 articles were excluded. Eventually, 37 articles

were included in the systematic review although only 27 provided sufficient data for meta-analysis (Fig. 1).

Displayed is the PRISMA (preferred reporting items for systematic reviews and meta-analyses) flow of search methodology and literature selection process.

33 of 37 studies utilized retrospective data, two used prospective data, and two more with both. 18 of 37 studies used publicly available data, among which the International Skin Imaging Collaboration (ISIC)

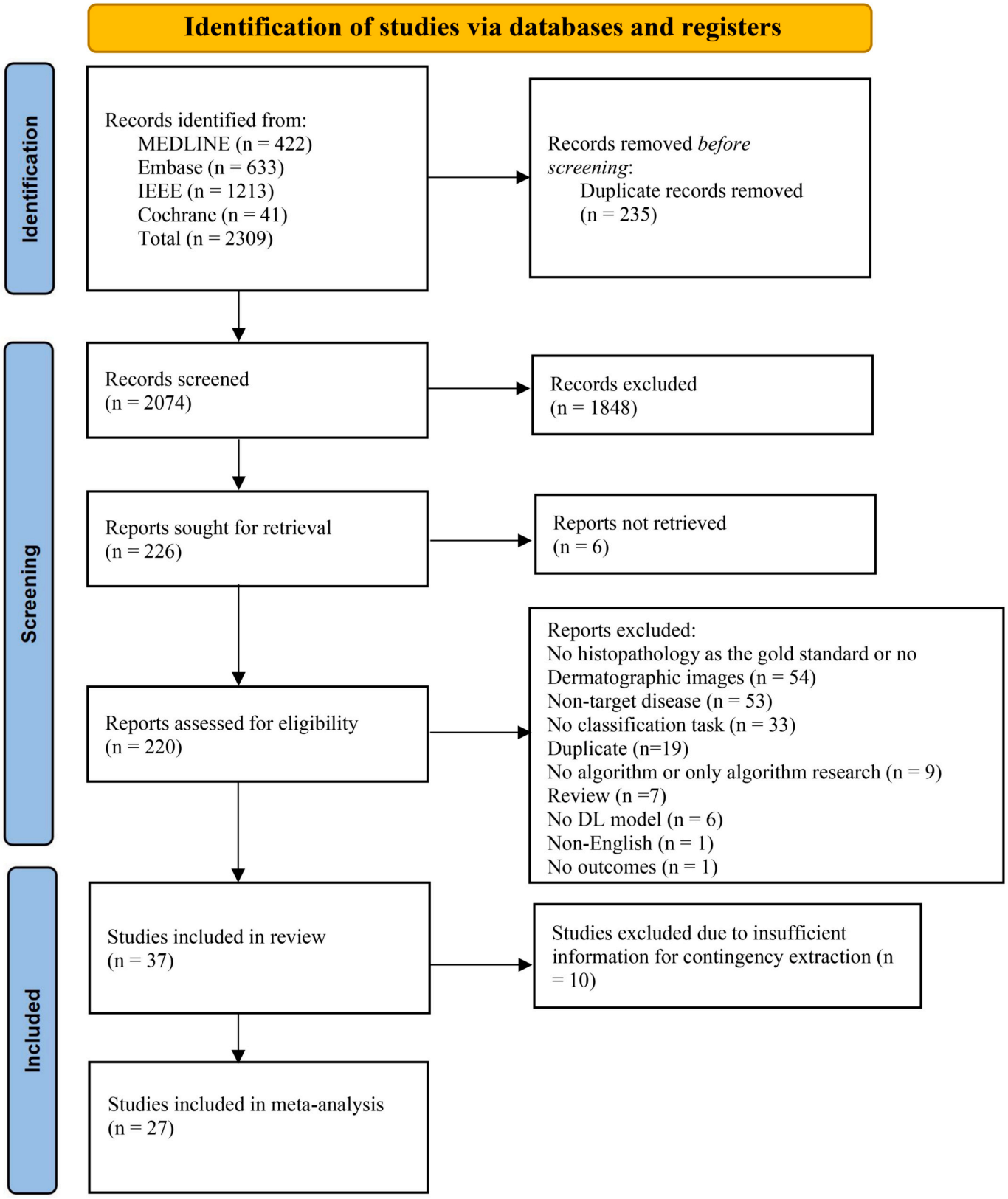


Fig. 1. PRISMA (2020) flowchart of study selection.

archive and Human Against Machine with 10,000 training images (HAM 10,000) were the most commonly accessed. 15 used restricted-access datasets (namely clinical or university-affiliated hospital data), and four further studies utilized two or more sources.

The exclusion of low-quality images was explicitly mentioned in eight studies. The criteria involved in determining image clarity involved artifacts such as hair, artificial color markings, surrounding tissue that were highly ulcerated, and those with/without substantial bleeding. Only four studies used out-of-sample datasets for external validation. 19 studies compared DL algorithms against human dermatologists in the same datasets, of which five have carried out DL-assisted dermatologists' research and seven studies compared with levels of seniority. Table 1 (below), and Table 5 as well as Table 6 in the Supplementary file 2 provide detailed characteristics of the included studies.

3.2. Pooled performance of DL algorithms

27 of 37 provided sufficient data to create a contingency table to calculate diagnostic performance. 20 studies provided data which was directly entered into the contingency table. Seven studies provided data which needed to be converted from categorical into binary data. The hierarchical SROC curves of these studies have been provided in Fig. 2A. When synthesizing data, the pooled SEN and SPE of all algorithms were 82 % (95 % CI 77–86 %) and 87 % (95 % CI 84–90 %) respectively. The AUC was 0.92 (95 % CI 0.89–0.94).

Many studies had used more than one DL algorithm or the same algorithm to report diagnostic performances over different datasets, therefore we listed DL algorithms with the highest diagnostic accuracy in each of these 27 studies. The pooled SEN and SPE were 83 % (75–89 %) and 89 % (84–93 %) respectively, and the AUC was 0.93 (95 % CI 0.90–0.95). Please refer to Fig. 2B for details.

3.3. Quality assessment

QUADAS-AI was used to evaluate the quality of all included studies (Fig. 1 in the Supplementary file 2). Detailed evaluations of each study are provided with an appropriate diagram in Fig. 2 of Supplementary file 2.

3.4. Subgroup meta-analyses

(1) Human-machine comparison based upon DL models versus dermatologists was conducted in 16 studies, and the DL algorithm was pooled for SEN and SPE of 79 % (95 % CI 66–87 %) and 81 % (95 % CI 74–87 %), with an AUC of 0.87 (95 % CI 0.84–0.90). Among the 16 studies, most were conducted using differential melanoma diagnosis according to different levels of seniority (i.e. junior or senior dermatologists) and types of doctors (i.e. dermatologists, physicians, or clinicians). The pooled SEN and SPE were 78 % (95 % CI 72–82 %) and 74 % (95 % CI 67–81 %) respectively, with an AUC of 0.83 (95%CI 0.79–0.86). Please see Fig. 3 for further details.

(2) Human-machine comparison according to level of seniority we conducted in only seven studies where DL algorithms were compared to both junior and senior dermatologists. The pooled DL algorithm had an SEN and SPE of 86 % (95 % CI 79–91 %) and 79 % (95 % CI 72–85 %) respectively, with an AUC of 0.90 (95 % CI 0.87–0.93). For junior dermatologists, a total of eight contingency tables were extracted, and the pooled SEN and SPE were 73 % (95 % CI 63–81 %) and 77 % (95 % CI 63–87 %), with the AUC of 0.80 (95 % CI 0.76–0.83). For senior dermatologists, the pooled SEN and SPE were 84 % (95 % CI 78–89 %) and 87 % (95 % CI 68–95 %), and the AUC was 0.88 (95 % CI 0.85–0.91). See Fig. 4 for further details.

(3) Human-machine cooperation with DL models versus unassisted dermatologists and DL-assisted dermatologists was conducted in only five studies. DL algorithms were pooled and had an SEN and SPE of 83 % (95 % CI 75–88 %) and 81 % (95 % CI 66–91 %) respectively, with an AUC of 0.88 (95 % CI 0.85–0.91). For unassisted dermatologists, a total of eight contingency tables were extracted. The pooled SEN and SPE were 75 % (95 % CI 62–85 %) and 68 % (95 % CI 59–75 %) respectively, and the AUC was 0.76 (95 % CI 0.72–0.79). For DL-assisted dermatologists, the SEN and SPE were 81 % (95 % CI 73–87 %) and 81 % (95 % CI 76–85 %), and the AUC was 0.87 (95 % CI 0.84–0.90). See Fig. 5 for further details.

3.5. Heterogeneity analysis

27 studies found that the AI algorithm was useful for diagnosing melanoma via dermatoscopic images under the random-effects model. The Spearman correlation coefficient was $r = 0.16$, ($P = 0.26$) which suggests there is no significant threshold effect. I^2 of SEN = 92.81 % ($P < 0.001$), while I^2 of SPE = 99.01 % ($P < 0.001$). Detailed results to subgroups and meta-regression analyses are provided in Table 2 and in Supplementary file 2, Figs. 4–8. The funnel plot generated suggests there is no publication bias ($P = 0.88$). See Supplementary file 2, Fig. 3 for details.

4. Discussion

This systematic review with meta-analysis assessed the performance of DL algorithms in diagnosing melanoma. The results indicate that DL algorithms perform well and are comparable to senior dermatologists. This finding was consistent when senior dermatologists independently detected melanoma and when dermatologists were AI-assisted. This study also presented the characteristics of the included studies in detail, and there were substantial differences. In addition, the quality of reports was rigorously assessed according to an adapted QUADAS-AI assessment tool. This enables us to discuss the performance of DL models in melanoma case detection, the relationship between human and machine, the characteristics of the most recent studies, and biases within this emerging evidence base.

This study demonstrated that DL shows promise in diagnosing melanoma, achieving an accuracy level comparable to that of dermatologists. Across the 27 studies meta-analyzed, we found that DL algorithms had slightly higher accuracies, with pooled sensitivity at 82 % (range: 77–86 %), specificity at 87 % (range: 84–90 %), and an AUC of 0.92 (range: 0.89–0.94). These findings are similar to previous studies [29,65]. Specifically, estimates of sensitivity associated with dermatologists were 70 % in the Nevisense trial [66] and 78 % in the MelaFind trial [67]. Other similar studies have also found dermatologists' diagnostic sensitivities of <80 % [68,69]. We found that the diagnostic accuracy of studies mainly ranged from 1 to 10 % although, several extreme values must also be considered. For example, Tschandl et al.'s' study had an extremely low sensitivity of 0.05 (range 0.01–0.17) [16]. This was perhaps because they utilized 9-classifications, and there were only 95 melanoma images (accounting for 0.69 %) in the training set and 40 melanoma images (i.e. 4.10 %) in the validation set. Similarly, the Marchetti et al. study found a very high specificity of 1.00 (range 0.99–1.00) but with a low sensitivity of 0.26 (0.17–0.38) [39]. This may be related to the operator setting the decision boundary threshold or because of the quality of the images used during training and testing [65,70]. Some studies may have also used penalty values to tradeoff specificity for improved sensitivity, or vice versa [65]. However, extreme values were unavoidable in this study and meta-analytical sensitivity analysis found little change in sensitivity and specificity after excluding the aforementioned studies. Therefore, DL performs well

Table 1
Model training, validation, and data sources.

First author, year	Target condition	Reference standard	Type of internal validation	External validation	Algorithm name	Source of data	Number of images for training/ validation/ testing	Open access data
Huang et al., 2021 [35]	Skin Cancer (melanoma)	Histopathology	Random split-sample validation	No	CNN	Retrospective study; data from (a) KCGMH data from the Department of Dermatology in KCGMH; (b) HAM10000.	9042/1130/ 1130	KCGMH data: No; HAM1000011: Yes
Iyatomi et al., 2008a [36]	Melanoma	Histopathology, Expert consensus	Leave-one-out cross-validation.	No	ANN	Retrospective study; data from four Japanese hospitals and two European university hospitals.	NR	No
Kwiatkowska et al., 2021 [37]	Melanoma	Histopathology	Random split-sample validation	No	CNN	Retrospective study; data from ISIC 2018: Skin Lesion Analysis Towards Melanoma Detection grand challenge—HAM10000 dataset.	8123/886/ 1006	Yes
Li et al., 2020 [38]	Skin Cancer (melanoma)	Histopathology	NR	No	CNN	Retrospective study; data from the China-Japan Friendship Hospital.	NR/NR/106	No
Marchetti et al., 2018 [39]	Melanoma	Histopathology, Expert consensus	Random split-sample validation	No	International Symposium on Biomedical Imaging challenge: twenty-five teams	Retrospective study; data from ISIC Archive.	900/NR/ 379	Yes
Maron et al., 2020 [19]	Melanoma and nevi	Histopathology	k-fold cross validation	No	CNN	Retrospective study; data from ISIC archive (with a large fraction of images coming from the HAM10000 dataset).	4894/NR/ 50	Yes
Marchetti et al., 2020 [24]	Melanoma	Histopathology	Random split-sample validation	No	Public challenge: twenty-three algorithms from twenty-three teams	Retrospective study; data from ISIC Archive (ISIC 2017).	2000/150 /600	Yes
Esteva et al., 2017 [40]	Skin Cancer (melanoma)	Histopathology	Nine fold cross validation	No	CNN(Net Inception V3)	Retrospective study; data from open-access dermatology repositories, the ISIC Dermoscopic Archive, the Edinburgh Dermofit Library, and data from the Stanford Hospital.	127,463/ 1942	Yes
Dascalu et al., 2019 [41]	Skin Cancer (melanoma)	Histopathology	Random split-sample validation	No	CNN (Inception V2)	Prospective study; data from ISIC 2017 and dermatology clinics at three sites (Maccabi Healthcare, Israel).	8:NR:2	ISIC 2017 Yes; dermatology clinics at three sites (73): Yes
Hoffmann et al., 2003 [42]	Skin Cancer (melanoma)	Histopathology, Expert consensus	NR	No	ANN	Retrospective study; data from Diagnostic and neural analysis of skin cancer study group, including 14 members from various European countries	NR	NR
Hekler et al., 2019 [43]	Skin Cancer (melanoma)	Histopathology, Expert consensus	k-fold cross validation	No	CNN (ResNet50)	Retrospective study; data from the ISIC archive, most images came from the HAM10000 Datase.	12,336/NR/ 300	Yes
Haenssle et al., 2018 [23]	Melanoma	Histopathology, follow-up	NR	No	CNN (Google's Inception V4)	Retrospective study; data from the high-quality validated image library of the Department of Dermatology, University of Heidelberg, Germany and the International Skin Imaging Collaboration (ISIC): the 2016 International Symposium on Biomedical Imaging (ISBI) challenge.	300/100/ NR/NR	Yes
Hosny et al., 2019 [44]	Melanoma and nevi	Histopathology	Ten fold cross validation	No	CNN (AlexNet)	Retrospective study; data from ISIC、MED-NODE and DermIS-DermQuest.	NR	Yes
Brinker et al., 2019a [45]	Melanoma	Histopathology, Expert consensus	NR	No	CNN (ReNet50)	Retrospective study; data from the International Skin Imaging Collaboration (ISIC) image archive.	12,378/ 1259/100	Yes

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Table 1 (*continued*)

First author, year	Target condition	Reference standard	Type of internal validation	External validation	Algorithm name	Source of data	Number of images for training/ validation/ testing	Open access data
Brinker et al., 2019b [46]	Melanoma	Histopathology, Expert consensus	NR	Yes	CNN (ReNet50)	Retrospective study; data from the International Skin Imaging Collaboration (ISIC) image archive and the HAM10000 dataset.	12,378/ 1359/100	Yes
Brinker et al., 2019c [47]	Melanoma	Histopathology	NR	No	CNN (ReNet50)	Retrospective study; data from the International Skin Imaging Collaboration image archive.	4024/NR/ 804	Yes
Dalal et al., 2011 [48]	Melanoma	Histopathology	Leave-one-out cross validation	No	ANN	Retrospective study; data from four private-practice clinics in Columbia, Missouri, Plantation, Florida, Rolla, Missouri, and Stamford, Connecticut.	75/NR/169	No
Mara et al., 2021 [49]	Melanoma	Histopathology, Expert consensus	Random split-sample validation	No	RCNN	Retrospective study; data from the ISIC2019 and PH2 datasets and the HIAE's teledermatology project.	NR	Yes
Gu et al., 2020 [50]	Skin Cancer (melanoma)	Histopathology, Expert consensus	Random split-sample validation	No	CNN, GAN	Retrospective study; data from two skin disease datasets MoleMap and HAM (HAM dataset is the largest publicly available skin dataset collected from ISIC archive).	NR	Yes
Hekler et al., 2020 [51]	Melanoma and nevi	Histopathology, Expert consensus	Four fold cross validation	No	CNN (ResNet50)	Retrospective study; data from database HAM10000 and the ISIC archive.	420/NR/ 384	Yes
Sies et al., 2020 [52]	Skin Cancer (melanoma)	Histopathology, Expert consensus, follow up	NR	No	CNN	Prospective study; data from real-world dermoscopic data that was prospectively acquired during 15 years of follow-up examinations (1997–2012) in 435 patients at increased melanoma risk.	NR	No
Piccolo et al., 2002 [53]	Melanoma	Histopathology	NR	No	ANN	Retrospective study; data from that were observed and subsequently excised over a 6 month period from 289 patients.	NR	No
Rubegni et al., 2002 [54]	Melanoma and nevi	Histopathology	Leave-one-out cross validation	No	ANN	Retrospective study; data from the Department of Dermatology of Siena University	NR	No
Stoecker et al., 2005 [55]	Melanoma	Histopathology	Leave-one-out cross validation	No	ANN	Retrospective study; data from Skin and Cancer Associates, Plantation, Florida, and Dermatology Associates of Tallahassee Florida.	NR	No
Khan et al., 2009 [56]	Melanoma	Histopathology	Random split-sample validation	No	ANN	Retrospective study; data from the EDRA Interactive Atlas of Dermoscopy, Skin and Cancer Associates, Plantation, Florida, and Dermatology Associates of Tallahassee, Florida.	382/NR/42	No
Pathan et al., 2018 [57]	Melanoma	Histopathology	Five fold cross validation, Random split-sample validation	No	ANN	Retrospective study; data from PH2 dataset.	NR	Yes
Yu et al., 2018 [21]	Melanoma	Histopathology	Two fold cross validation	No	CNN	Retrospective study; data from the Severance Hospital in the Yonsei University Health System, Seoul, Korea, and the Dongsan Hospital in the Keimyung University Health System, Daegu, Korea.	253/109/ 362	
Maron et al., 2019 [58]	Skin Cancer (melanoma)	Histopathology, Expert consensus	NR	No	CNN (ResNet50)	Retrospective study; data from the International Skin Imaging Collaboration (ISIC) archive; most images came from the HAM10000 Dataset.	12,336/NR/ 300	Yes

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Table 1 (*continued*)

First author, year	Target condition	Reference standard	Type of internal validation	External validation	Algorithm name	Source of data	Number of images for training/ validation/ testing	Open access data
Tschandl et al., 2019a [16]	Skin Cancer (melanoma)	Histopathology	NR	Yes	CNN	Retrospective study; data from a primary skin cancer clinic in Queensland, Australia, and The test set originated from multiple sources, including the Medical University of Vienna.	13,724/ 975/2072	No
Walker et al., 2019 [59]	Skin Cancer (melanoma)	Histopathology	NR	Yes	CNN (Inception V2)	Retrospective study: data from the International Skin Imaging Collaboration (ISIC) 2017 dataset, and the Interactive Atlas of Dermoscopy (IAD) dataset; Prospective study: dermatologic clinic (AD, Tel Aviv, IL), Israel.	3954/NR/ 545	ISIC2017:Yes ; clinical database: No
Warsi et al., 2019 [60]	Melanoma	Histopathology	Random split-sample validation	No	ANN	Retrospective study; data from PH2 database.	140/30/30	Yes
Lee et al., 2020 [22]	Melanoma	Histopathology	NR	No	DCNN	Retrospective study; data from the Department of Dermatology, Severance Hospital, Seoul, Korea.	872/NR/ 200	No
Winkler et al., 2020 [61]	Melanoma	Histopathology, follow up	NR	No	CNN	Retrospective study; data from three image libraries (Departments of Dermatology, Universities of Heidelberg, Munich and Lyon).	NR/NR/130	No
Minagawa et al., 2021 [20]	Skin Cancer (melanoma)	Histopathology, Expert consensus	NR	No	DNN	Retrospective study; data from the ISIC 2017 dataset, HAM10000 dataset, BCN20000 dataset (ISIC set), and Shinshu set.	12,785/NR/ 700	Shinshu set: No ; others: Yes
Tschandl et al., 2019 [62]	Skin Cancer (melanoma)	Histopathology, Expert consensus	Random split-sample validation	Yes	Public challenge:139 algorithms	Retrospective study; data from the ISIC 2017 dataset, HAM10000 dataset, BCN20000 dataset (ISIC set), and Shinshu set of Japanese patients who were seen in the Department of Dermatology at Shinshu University Hospital between January 2008 and December 2017.	100,015/ NR/1511	Yes
Dreiseitl et al., 2009 [63]	Melanoma	Histopathology, Expert consensus, follow up	NR	No	ANN	Prospective study; data from the Department of Dermatology of the Medical University of Vienna.	1311/NR/ 3021	No
Iyatomi et al., 2008b [64]	Melanoma	Histopathology, Expert consensus, follow up	Leave-one-out cross validation	No	ANN	Retrospective study; data from four university hospitals (University of Naples, Italy; University of Graz, Austria; Vienna University, Austria; Keio University, Tokyo).	NR	No

Abbreviations: NR = not reported, CNN = convolutional neural network, KCGMH = Kaohsiung Chang Gung Memorial Hospital, HAM10000 = Human Against Machine with 10,000 training images, ANN = artificial neural network, ISIC = International Skin Imaging Collaboration, VGG16 = Visual Geometry Group Network, HIAE = Hospital Israelita Albert Einstein, RCNN = Region Convolutional Neural Network, GAN = generative adversarial networks, BPNN = backpropagation neural network, DCNN = deep CNN; ResNet50 = Residual network 50.

in diagnosing melanoma but it remains necessary to compare the diagnostic abilities of DL versus dermatologists with the same big dataset.

4.1. Human-machine comparison

DL algorithms appear to have a superior diagnostic performance in sensitivity, specificity, and AUC, which were all consistent with similar studies [28,45]. In a similar study, Haggenmüller et al. systematically reviewed studies to assess the diagnostic performance of CNN using melanoma images (i.e., dermoscopic images, clinical images, and histopathological images). They found CNN is superior or at least equivalent compared to physicians when diagnosing melanoma [27]. This suggests that DL can achieve and may even surpass the human standard soon. Furthermore, comparisons of dermatologists according to levels of seniority suggest that DL is significantly better than less experienced dermatologists and at least equivalent to senior dermatologists. This achievement has primarily been attributed to CNNs that exhibit specific responses to visual stimuli [15,16]. During the image processing stage, CNNs break down images into pixels and efficiently process this information before extracting features and generating predictions. This capability enables DL to better understand and analyze medical images. Therefore, it is reasonable to suggest that as algorithms are fed more data, they will improve even beyond the capabilities of senior dermatologists.

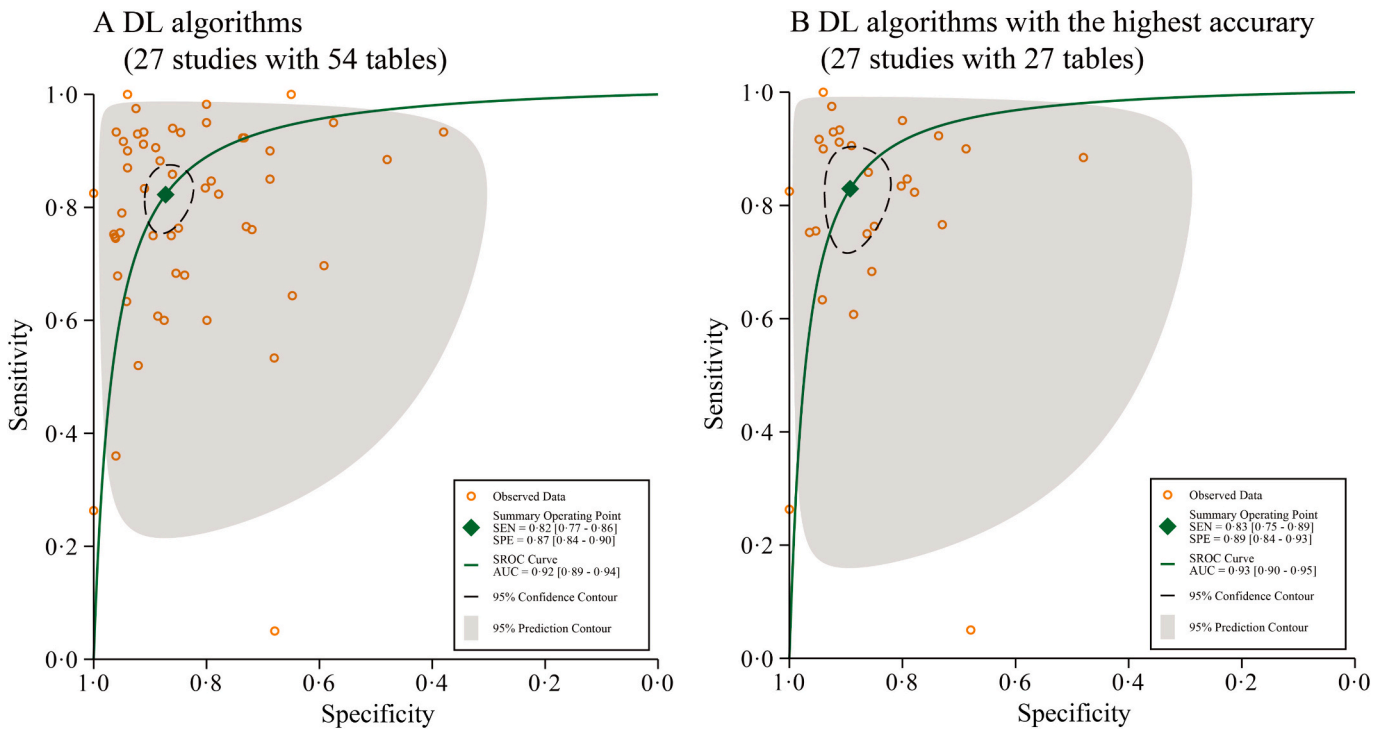


Fig. 2. Hierarchical ROC curves of studies for DL model using different data sources for comparing performance between binary and categorical datasets. A, Receiver operator characteristic (ROC) curves of all studies included in the meta-analysis (27 studies with 54 tables); B, ROC curves of studies reporting the highest accuracy (27 studies with 27 tables).

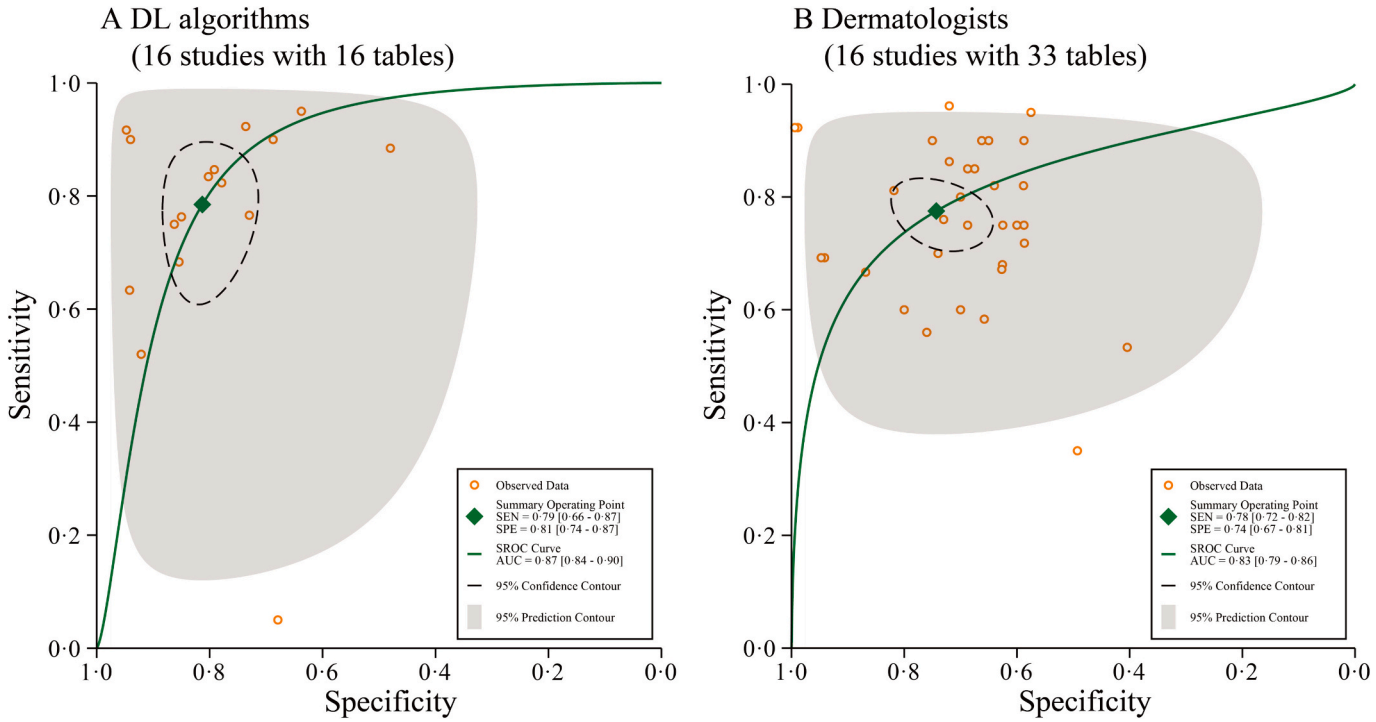


Fig. 3. Hierarchical summary receiver operating characteristic curves of studies for human-machine comparison with DL models versus dermatologists. A, Receiver operator characteristic (ROC) curves of studies for DL model reporting the highest accuracy (16 studies with 16 tables); B, ROC curves of studies for all dermatologists (16 studies with 33 tables).

4.2. Human-machine cooperation

The diagnostic performance of DL-assisted dermatologists appeared no different from DL algorithms alone; however, DL-assistance significantly improved dermatologists' alone. Evidence also supports this finding and suggests that DL-assistance can significantly advance diagnostic performance overall and highlights the auxiliary effect of DL algorithms [71,72]. Notably, there was a significant difference in

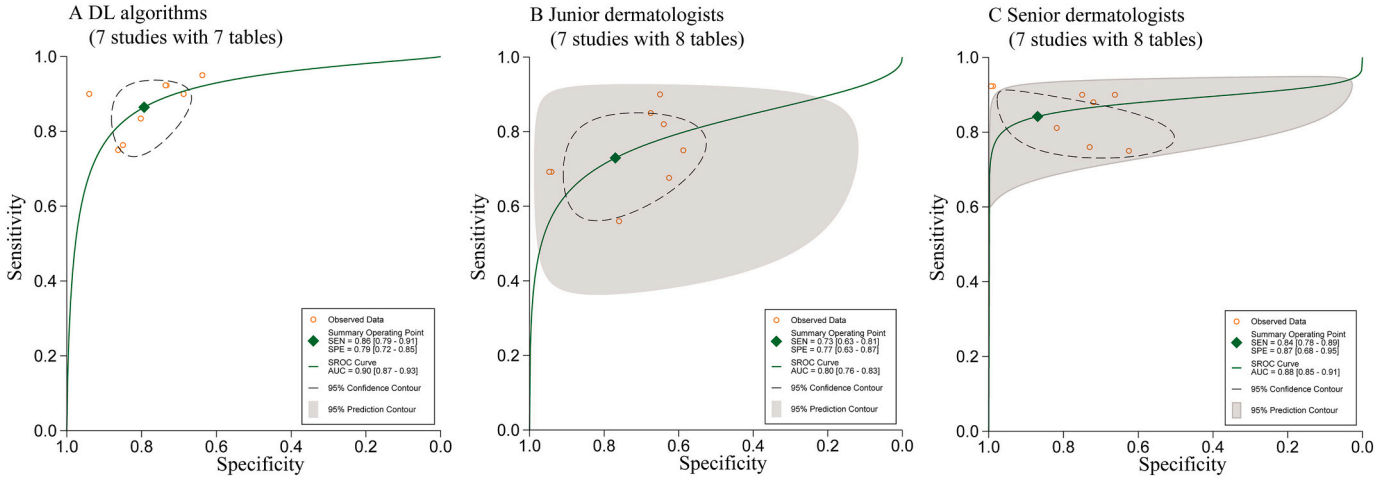


Fig. 4. Hierarchical summary receiver operating characteristic curves of studies for human-machine comparison with different seniority.
A, Receiver operator characteristic (ROC) curves of studies for DL model reporting the highest accuracy (7 studies with 7 tables); B, ROC curves of studies for all junior dermatologists (7 studies with 8 tables); C, ROC curves of studies for all senior dermatologists (7 studies with 8 tables).

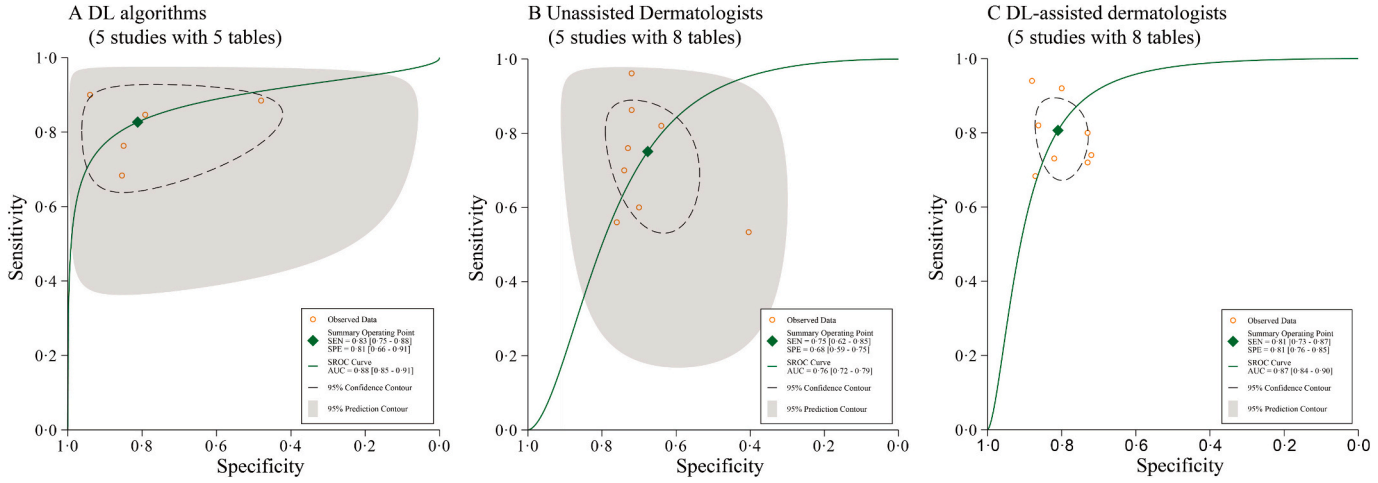


Fig. 5. Hierarchical summary receiver operating characteristic curves of studies for human-machine cooperation with DL models versus unassisted dermatologists and DL-assisted dermatologists.
A, Receiver operator characteristic (ROC) curves of studies for DL model reporting the highest accuracy (5 studies with 5 tables); B, ROC curves of studies for unassisted dermatologists (5 studies with 8 tables); C, ROC curves of studies for DL-assisted dermatologists (5 studies with 8 tables).

specificity between unassisted and DL-assisted dermatologists. Therefore, we suggest that the discrepancy in AUCs can be explained by specificity, indicating that DL algorithms improved dermatologists' ability to diagnose non-melanoma cases. This may be due to the high mortality rate associated with melanoma, as well as dermatologists' fear of missed diagnoses, as well as the difficulty distinguishing achromatic melanoma from other skin lesions [16]. However, with the help of AI, the diagnostic performance of dermatologists improved in both sensitivity and specificity, overall. Having this support mechanism may also assure clinicians about their decisions. Roman et al. [19] also concluded that the dermatologists' levels in decision confidence increase with CNN-based confirmation. Therefore, we are convinced that the DL algorithms will become an effective auxiliary tool to help dermatologists identify melanoma cases.

AI/DL algorithms perform well in diagnosing melanoma, but can they be used independently? This study suggests that AI should not be used to diagnose melanoma independently. This is despite multiple studies showing that pathologists have high inter- and intra-observer variability in the diagnosis of melanocytic tumors, especially for those with ambiguous histologic features [73,74]. Elmore et al. [75] also showed the divergence rate among different pathologists is up to 75 % when trying to distinguish moderate to severe dysplasia nevus from early melanoma. As the gold standard of diagnosis, pathology seems to be the best choice especially because AI still generates errors which could lead to missed opportunities, misdiagnoses, and unnecessary biopsies which have economic and psychological implications for both patients and practitioners. AI/DL also uses all kinds of static dermoscopic images or clinical photos when diagnosing skin cancer. Static images do not necessarily represent all clinical settings therefore diagnostic performances may be underestimated for AI-assisted dermatologists.

In practice, skin tumors are not classified into binary categories therefore DL will most likely misdiagnose lesions without nuanced foundational training. Therefore, we need to consider the differences between the image characteristics used for algorithm development and those that might be experienced in the real world. In addition, disease diagnosis is a two-way process where patients and doctors interact. And, some have suggested that relying on mathematical models alone is tantamount to ignoring human existence [76]. Dermatology relies on tactile examination, as some melanomas may become more noticeable

Table 2
Summary estimate of pooled performance of artificial intelligence in image-based melanoma detection.

	No. of studies	No. of contingency tables	Sensitivity			Specificity			r_s	*P*–value[b]
			SEN (95%CI)	*P*–value[a]	I^2 (95%CI)	SPE (95%CI)	*P*–value[a]	I^2 (95%CI)		
Overall for DL algorithms	27	54	0.82(0.77–0.86)	<0.001	92.81(91.49–94.13)	0.87(0.84–0.90)	<0.001	99.01(98.91–99.10)	0.16	0.26
The highest Acc of DL algorithms	27	27	0.83(0.75–0.89)	<0.001	94.44(93.10–95.79)	0.89(0.84–0.93)	<0.001	99.49(99.44–99.55)	0.02	0.91
DL models versus dermatologists										
DL algorithms	16	16	0.79(0.66–0.87)	<0.001	92.88(90.44–95.31)	0.81(0.74–0.87)	<0.001	99.31(99.19–99.42)	−0.11	0.09
Dermatologists	16	33	0.78(0.72–0.82)	<0.001	81.52(75.69–87.34)	0.74(0.67–0.81)	<0.001	96.44(95.75–97.12)	−0.05	0.70
DL models versus dermatologists with different seniority										
DL algorithms	7	8	0.86(0.79–0.91)	0.07	45.93(1.99–89.87)	0.79(0.72–0.85)	<0.001	87.41(80.04–94.78)	0.71	0.048
Junior dermatologists	7	8	0.73(0.63–0.81)	<0.001	79.06(65.01–93.10)	0.77(0.63–0.87)	<0.001	95.52(93.59–97.46)	0.23	0.59
Senior dermatologists	7	8	0.84(0.78–0.98)	<0.001	76.40(60.09–92.70)	0.87(0.68–0.95)	<0.001	96.29(94.77–97.80)	−0.68	0.07
DL models versus dermatologists and DL-assisted dermatologists.										
DL algorithms	5	5	0.83(0.75–0.88)	<0.001	78.34(59.41–97.28)	0.81(0.66–0.91)	<0.001	99.77(99.72–99.82)	0.00	>0.99
Unassisted Dermatologists	5	8	0.75(0.62–0.85)	<0.001	84.49(74.88–94.10)	0.68(0.59–0.75)	<0.001	94.31(91.67–96.96)	−0.05	0.91
DL-assisted dermatologists	5	8	0.81(0.73–0.87)	0.01	64.51(37.57–91.45)	0.81(0.76–0.85)	<0.001	65.63(39.73–91.54)	−0.23	0.59

r_s = spearman correlation coefficient, DL = deep learning, Acc = accuracy, SEN = sensitivity, SPE = specificity.
[a] P–value for I^2 to test the heterogeneity of non-threshold effect.
[b] P–value for spearman correlation coefficient to test the heterogeneity of threshold effect.

with touch [7,77]. This is why it is particularly important for dermatologists to make judgments based on a number of different sources such as clinical manifestations, dermatoscopy, histology, and pathology. They can also consult experienced clinicians if uncertain therefore DL should not yet be used to make clinical decisions. Clearly, there are a number of challenges to address before we even consider independent DL-based diagnostics.

While DL algorithms have improved in diagnosing dermoscopic images, there is currently insufficient evidence to support their independent use in even the most common cases. Of course, this field of technology is still evolving and requires further empirical research and validation, and even though DeepX Diagnostics' DermoSight received FDA approval in 2023 for remote skin cancer screening of suspicious lesions, it still primarily serves as an auxiliary tool rather than an independent diagnostic device [78]. This tells us that DL algorithms require additional training and development to recognize borderline cases which tend to be early-stage melanoma. Therefore, DL is at present an auxiliary diagnostic tool for dermatologists but there are several necessary auxiliary functions. For example, DL may be embedded into triage, and screening, or might provide diagnostic references, thereby reducing practitioners' and physicians' workloads and enhancing diagnostic efficiencies and accuracies. However, exploring optimal workflows and processes in relationship between DL technologies and dermatologists is also required to achieve more accurate and efficient diagnostic pathways.

4.3. Limitations of included studies

The construction and development of DL models must be based on big data. Only through training and validation with a sufficient number of images can the diagnostic efficiency of these models be improved. Unfortunately, this study included few dermatoscopic images. The number of images in general training models was also only several thousand and some of the included articles used only several hundred images as training sets. Additionally, the number of melanoma images was between 1/3–1/8 of the non-melanoma images in this study, overall. This may have led to unstable diagnostic performances and also a decrease in model sensitivity. The low number of images was mainly due to the fact that melanoma accounts for only 4 % of all skin cancers [79], combined with dermoscopic images and ethical restrictions. Furthermore, the generalizability of the model is an important index which is a prerequisite for the promotion and application of the model. Only four studies in this review used external validation sets and 89 % used only internal validation procedures. Additionally, 1/3 of this sample used data augmentation to expand datasets but this did not change the evident lack of data which might have also caused an overestimation in performances which again limits generalizability [80,81]. We suggest that research institutions strengthen inter-organizational cooperation and establish a cloud platform to make full use of common image resources, and use an increased number and higher quality samples to develop models.

The included studies were mainly retrospective, which may be related to the accessibility of electronic medical records and online open-access databases. It should also be noted that image data was collated retrospectively and reused for training which may not be representative of diagnoses or typical characteristics [70]. We should use an easily obtainable retrospective data set to test whether AI\DL algorithms are appropriate. This will help us to modify and improve the accuracy of these technologies for cancer case identification. The Checklist for Evaluation of Image-Based Artificial Intelligence Reports in Dermatology (CLEAR Derm) consensus guidelines [82] also point out that preliminary evaluation does not need to take the form of prospective clinical trials but can use retrospective data to explore value and identify any early concerns before larger prospective clinical trials are commenced.

Most research into DL-based diagnostic models uses publicly

available datasets for the development of DL algorithms. This enables researchers to efficiently collect data [83] and to establish benchmarks to compare algorithm performances [84,85]. However, there are also some drawbacks to this approach, such as the limited number of characteristics provided to analyze and nonstandard inclusion/exclusion criteria. The results of a study of publicly available skin cancer image datasets by Wen et al. showed that there are differences in the relation to country of origin, year of dataset publication, the completeness of datasets, ethical approval, informed consent, skin images acquisition methods, and the number of skin tumors contained within data sets [86]. More specifically, data characteristics and metadata are limited, with substantial under-representation of darker skin types [83,86]. This will have influenced the performance of algorithms due to the lack of high-quality images and metadata images, as well as types of lesions. Many studies would have also excluded fuzzy images which captured hair, blood vessels, air bubbles, etc. which would have affected training. While this may have improved the results of model training, the diagnostic performance will have been overestimated. This is a concern because it is unclear whether the included datasets contained an appropriate representation of the population. It is hoped that standards will be established for publicly available datasets to ensure that more high-quality metadata images become accessible to scientific researchers.

The interpretability of the algorithm also requires further attention. At present, to obtain high diagnostic performance, DL may use a large number of complex parameters with multiple hidden layers. This means we do not know how DL can draw specific conclusions using imputed data. We also do not know when DL models will be mistaken, and if so, who will bear the responsibility for errors. The European Union's General Data Protection Requirements stipulate explainability as a requirement of algorithm decision-making [87,88]. The Food and Drug Administration recommends that until there is substantial clinical trial evidence, AI should only assist in decision-making rather than making autonomous decisions, with clinicians maintaining responsibility [89]. Explaining AI has been studied to overcome the issues caused by this so-called 'black box' [90]. Compared to AI technologies, explainable AI (XAI) can provide decision-making and model explanations [91]. However, the hope is that big data scientists and clinicians will develop XAI together although this presents some additional complexities which must also be explored.

Since AI is a relatively new discipline, traditional statements, guidelines, and standards do not fully apply to AI-based medical research. Presently, the QUADAS-2 tool is being widely used to assess bias and the applicability of diagnostic test accuracy (DTA) [31]. However, evidence suggests that 94 % of the published AI-based diagnostic test accuracy (AI DTA) systematic reviews are conducted without an AI-specific quality assessment tool [92]. The QUADAS-2 tool does not factor in AI DTA quality evaluation and does not intercalate professional terminologies or patient selection in AI DTA studies [33]. To tackle these potential sources of bias, we refer to the adjusted QUADAS-AI assessment tool and suggest recombination of the QUADAS-2 and QUADAS-C tools. Although again, this will require further research and interdisciplinary cooperation to develop an appropriate tool.

4.4. Strengths, limitations, and outlook

This was a rigorously conducted systematic review with meta-analysis, adhering to the PRISMA 2020 guidelines, comprehensively evaluating the performance of DL in diagnosing melanoma. Detailed analyses of human-machine comparison and cooperation were performed, highlighting the potential and importance of DL in medical diagnostics. This study also provides robust evidence in relation to the application of DL in diagnosing melanoma with dermatoscopic images. However, there were a number of limitations which should also be discussed. For instance, a subset of databases was retrieved in this study which might have excluded some related AI/DL research. In addition, the diagnostic performance of DL algorithms was compared to that of dermatologists according to levels of seniority but these studies did not all specify their experience-based thresholds. Therefore, we grouped data qualitatively according to the terms given i.e. 'junior' and 'senior'. This introduced an element of subjectivity and will have caused some classification bias. Another limitation is that this study focused on melanoma and non-melanoma, but the included studies did not all aim to distinguish melanomas. Therefore, again we extracted and had to combine data related to melanoma which may have reduced the precision of findings. Finally, we emphasize that only by conducting high-quality research can we get correct results given the current soaring AI/DL research. A checklist of items (CLEAR Derm consensus guidelines) was developed outlining best practices of image-based AI development and assessment in dermatology [82]. This will help both developers and reviewers achieve this goal but it is hoped that more high-quality research is also published to create a more reliable evidence base.

5. Conclusions

DL algorithms derived using dermoscopic images have shown favorable performances in diagnosing melanoma. This study also found that DL algorithms are as good as senior dermatologists in diagnosing melanoma but that DL assistance can also promote diagnostic performance of dermatologists. Although, this requires high-quality and large-scale multicenter research to confirm and address specific challenges faced by AI in medical diagnosis.

Supplementary data (Supplementary file 1: PRISMA 2020; Supplementary file 2: supplementary figures & tablesfile 1:) to this article can be found online at https://doi.org/10.1016/j.artmed.2024.102934.

Abbreviations

AI	artificial intelligence
CI	confidence intervals
DL	deep learning
HSROC	hierarchical summary receiver operating characteristic
SEN	sensitivity
SPE	specificity
AUC	area under the curve
PRISMA	preferred reporting items for systematic reviews and meta-analyses
TP	true-positive
FP	false-positive
TP	true-negative
FN	false-negative
QUADAS-2	quality assessment of diagnostic accuracy studies 2
QUADAS-C	QUADAS-Comparative
QUADAS-AI	artificial intelligence-centered diagnostic test accuracy studies
CNN	convolutional neural networks
OOD	out-of-distribution
DTA	diagnostic test accuracy

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Ethics statement

Not applicable.

Patient consent

Not applicable.

CRediT authorship contribution statement

Zichen Ye: Writing – original draft, Visualization, Software, Project administration, Methodology, Formal analysis, Data curation. **Daqian Zhang:** Formal analysis, Data curation. **Yuankai Zhao:** Formal analysis, Data curation. **Mingyang Chen:** Data curation. **Huike Wang:** Data curation. **Samuel Seery:** Writing – review & editing. **Yimin Qu:** Data curation. **Peng Xue:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Yu Jiang:** Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data generated or analyzed during this study are included in this published article and its supplementary information files. The original data can be obtained by the reasonable application of the corresponding author.

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